

Question: 1

Use the ping command to check the connectivity between your computer and www.spotify.com. Note the response times and explain what they indicate about your network connection?

Answer: . Ping command: ping www.spotify.com

- . It checks whether your computer can connect to Spotify's server.
- . A reply means the connection is working.
- . Response time (ms) shows how fast the server responds.
- . Lower response time means a faster and more stable connection.
- . High response time means the network may be slow or have delay.
- . Request timed out means there may be a connectivity problem.

Question: 2

Your friend is unable to access www.zomato.com, but other websites work fine. Use nslookup to check the DNS resolution for www.zomato.com and describe what the output tells you.

Hint: Look for the IP address and any error messages in the nslookup result.?

Answer: • Purpose of nslookup:

- Checks if the DNS server can convert the website name (www.zomato.com) into an IP address.
- Key components of the output:
- Server & Address: Shows the local DNS server name and its IP address being used.

- Non-authoritative answer: Indicates the result is fetched from a cached DNS record rather than the official domain server.
- Name: Displays the target website domain (www.zomato.com).
- Addresses: Lists the IPv4 or IPv6 addresses assigned to Zomato's servers.
- What the output tells us (Analysis):
- Scenario 1: Success (IP addresses are listed)
- The DNS resolution is working perfectly.
- The issue is likely a local browser issue, firewall block, or server-side downtime for that specific region.
- Scenario 2: Failure (Error message appears)
- If it says "DNS request timed out", the local DNS server is not responding.
- If it says "Non-existent domain (NXDOMAIN)", the DNS server cannot find the website record.

Question: 3

Open Wireshark and capture network traffic while you refresh your Instagram feed in a browser. Identify and list the different protocols you see in the capture.

Hint: Filter by protocol type (like HTTP, TCP, DNS) in Wireshark to make it easier?

Answer: • DNS (Domain Name System):

- Role: Resolves the domain name ://instagram.com into its corresponding IP address before communication begins.
- TCP (Transmission Control Protocol):

- Role: Handles the reliable transfer of data packets. You will see TCP handshakes (SYN, SYN-ACK, ACK) establishing connection channels between your browser and Instagram's servers.
- TLS (Transport Layer Security):
- Role: Encrypts the communication. Since Instagram uses secure HTTPS, the actual HTTP data is wrapped inside TLS packets to keep your login session and feed data private.
- HTTP/2 or HTTP/3:
- Role: The application layer protocol used to load the website content, feed text, and API data. (Note: Modern browsers often label HTTP/3 traffic under the QUIC protocol over UDP).
- UDP (User Datagram Protocol) / QUIC:
- Role: Used for fast media streaming. Instagram loads high-resolution images, Reels, and videos rapidly using QUIC/UDP to minimize latency.
- ARP (Address Resolution Protocol):
- Role: Operates in the background to map your local IP address to your router's physical MAC address so the packets can leave your device.

Question: 4

A user reports that their Flipkart app is slow to load product images. Use netstat to list all current network connections on your system and identify if there are any suspicious or high-usage connections that could be affecting bandwidth?

Answer: • Command to run:

- Use netstat -ano (Windows) or netstat -apn (Linux/macOS).
- -a: Lists all active connections and listening ports.
- -n: Displays addresses and port numbers in numerical form (speeds up the output).

- -o: Displays the PID (Process ID) associated with each connection (crucial for finding the high-usage app).
- How to identify high-usage or suspicious connections:
- ESTABLISHED State: Look for connections that show ESTABLISHED under the "State" column, meaning data is currently being transferred.
- Foreign Address (IPs): Check for unusual or unfamiliar remote IP addresses that are holding long, active connections.
- Repeated Connections: Multiple connections to the same foreign IP address often indicate a large download/upload or media streaming activity.
- Tracking the source application via PID:
- Note down the PID number of any connection that looks suspicious or has multiple active links.
- Open Task Manager (Ctrl + Shift + Esc) on Windows, go to the "Details" tab, and match the PID to see exactly which application or background process is consuming the network bandwidth.

Question: 5

Your college WiFi keeps disconnecting during IPL match streaming. Create a troubleshooting checklist using ping, tracert, and netstat to systematically diagnose the issue step by step. For each tool, write what command you would use and what result would indicate a problem?

Answer: LStep 1: Using ping (To check basic connectivity and packet loss) • Command: ping -t 8.8.8.8 (or ping -t google.com) • (Note: The -t flag pings continuously, which is useful during an active streaming drop). • Problem Indicator: • If the output shows "Request timed out" or a high percentage of "Packet loss", it means the connection is dropping packets. • Extremely high "time=..." values (e.g., >200ms) indicate severe network lag/congestion caused by too many users

streaming the IPL match at once. Step 2: Using tracert (To identify where the connection breaks) • Command: tracert 8.8.8.8 (Windows) or traceroute 8.8.8.8 (Mac/Linux) • Problem Indicator: • If the delay (high ms) starts at the very first hop (e.g., 192.168.1.1), the college Wi-Fi router/Access Point itself is overloaded. • If request timeouts * * * happen further down the path, the problem lies with the college's main Internet Service Provider (ISP) backbone. Step 3: Using netstat (To check for connection limits or socket exhaustion) • Command: netstat -ano • Problem Indicator: • Look for an unusually high number of active rows showing the SYN_SENT state. This indicates your system is trying to open connections, but the overloaded college network is not responding. • A massive list of connections originating from your browser or streaming app might mean local socket saturation.